

Request for Application

Application to Test in the NRIC DOME Facility

1. INTRODUCTION

Battelle Energy Alliance, LLC (BEA) is the Management and Operating Contractor of the U.S. Department of Energy (DOE) owned Idaho National Laboratory (INL). BEA invites industry applicants to collaborate with the National Reactor Innovation Center (NRIC) and INL to apply for the opportunity to perform advanced reactor experiments in the Demonstration of Microreactor Experiments (DOME) test bed facility beginning in July 2027. The facility is located at the INL's Material and Fuels Complex (MFC). This opportunity aims to accelerate the development and commercialization of advanced reactors through access and testing in a nuclear reactor test bed.

2. BACKGROUND

Advanced reactor concepts are being developed by multiple commercial entities. First-of-a-kind nuclear technology needs a safe, reliable, and affordable location for testing. It is difficult to find these facilities outside of the national laboratory system and it is cost prohibitive for individual reactor developers to build test facilities themselves. NRIC was established to “enable the testing and demonstration of reactor concepts to be proposed and funded, in whole or in part, by the private sector.” Sec. 958 of NEICA 2017.

NRIC is developing facilities and processes to allow advanced reactor developers to expedite commercialization by providing the infrastructure to host fueled reactor experiments developed by industry. NRIC/INL is seeking applications from industry applicants to schedule the DOME facility for fueled reactor experiment testing when the facility is ready for use after the first reactor experiment is complete, projected July 2027. The applications will include detailed information in the following general categories, each of which is a separate section of the application:

1. Reactor Technology Maturity (plan to mature technology prior to testing)
2. Fuel (plan to qualify, procure, deliver, store, and dispose)
3. Testing (tests planned in DOME)
4. Project Schedule (proposed schedule from application through decommissioning)
5. DOME Integration (special interface needs and clear understanding of reactor to DOME interface and shielding)
6. DOE Authorization Plan (plan for obtaining DOE authorization for reactor startup)
7. Funding (amounts and sources)
8. Experience and Capabilities (key personnel and partnerships, management infrastructure)

Attachment 1 of the application is a Master Document and Deliverable List (MDDL), which is to be submitted with the rest of the application. Attachment 2 of the submitted application will be the applicant's project schedule.

Detailed instructions, description of application information, and evaluation criteria are described in INL guide GDE-55115 DOME Scheduling Application (attached). Development of the application is the financial responsibility of the applicant.

3. ELIGIBILITY REQUIREMENTS

Testing in DOME is expected to be self-funded by the applicant, and applicants must meet the following criteria to be eligible:

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1. Applicants must demonstrate a level of project maturity commensurate with significant progress through conceptual design of the proposed reactor and DOE agreement or engagement on the proposed reactor Nuclear Safety Design Agreement (NSDA) or 10 CFR 830 safety basis equivalent.
2. Applicants must be:
 - a. United States-owned companies or pass a DOE-P 485.1A review, or
 - b. Companies incorporated or organized under the laws of any state with a parent company that:
 - i. Affords opportunities to United States-owned companies comparable to those afforded to any other company in joint ventures authorized under applicable law,
 - ii. Affords local investment opportunities comparable to those afforded to any other company, and
 - iii. Affords adequate and effective protection for intellectual property rights of United States-owned companies.

4. PROCESS

Responses to this opportunity notice will be considered as applications and will be evaluated by NRIC and DOE. The decision to schedule the DOME facility for an advanced reactor experiment will rely on the assessments of qualified applications. Only applicants that propose to test a fueled reactor experiment that meets the limitations/specifications of NRIC's DOME facility with a completed DOME scheduling application will be considered.

This opportunity does not guarantee selection of an applicant for testing in DOME or guarantee a specific time frame or priority for testing. Applications will be evaluated by INL and DOE as defined in GDE-55115 DOME Scheduling Application. NRIC will make a testing sequence recommendation to DOE that optimizes the use of DOME. DOE will determine the schedule of experiments in DOME. NRIC will communicate the results of the scheduling process to the applicants once the testing schedule is determined. Results are anticipated within four to six months of receipt of the applications.

5. APPLICATION REQUIREMENTS

Interested applicants must provide contact information consisting of company name, contact name, telephone number and email via email to the contacts listed below by 5:00 p.m. MDT on June 19, 2026. Respondents will receive additional instructions and a secure link via email for submitting applications. Applications are due by 5:00 p.m. MDT on July 8, 2026. Applications must include sufficient written information confirming the responder's qualifications and addressing each of the opportunity evaluation criteria listed in GDE-55115.

Applications must be no more than fifty (50) pages in length in PDF format, plus an attached MDDL and schedule. The MDDL and schedule do not count towards the 50 page limit. The PDF file shall include Table 1 and the sections in Table 2 from GDE-55115. Answers to the sections in Table 2 should include all assumptions made and bases to support these assumptions.

If you have questions, or need additional information, please contact Sam Reiss at Samuel.Reiss@inl.gov or AnnMarie Marshall at Ann.Marshall@inl.gov. Technical questions must be submitted in writing via email a minimum of two weeks prior to the closing of the opportunity to the contacts listed above. Responses to questions will be genericized and made available to all applicants through email.

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6. PROPRIETARY INFORMATION

If the information provided in response to this opportunity is business sensitive or proprietary to the responding party, respondents must mark information in their application accordingly and ensure there is an executed Non-Disclosure Agreement or similar agreement in place with the INL prior to submittal.